Math 534H - Spring 2026 - Lecture 8

Cont. from last time

$$\underbrace{5y}\, u_{xx} - 8u_{xy} + \underbrace{2x}\, u_{yy} = 0$$

$$\Delta = (5y)(2x) - (-4)^2 = \underbrace{10xy - 8}$$

When is $\Delta > 0$, $\Delta < 0$, $\Delta = 0$?

$$\Delta = 10xy - 8 > 0 \iff 5xy - 4 > 0 \iff \boxed{xy > \frac{4}{5}}$$

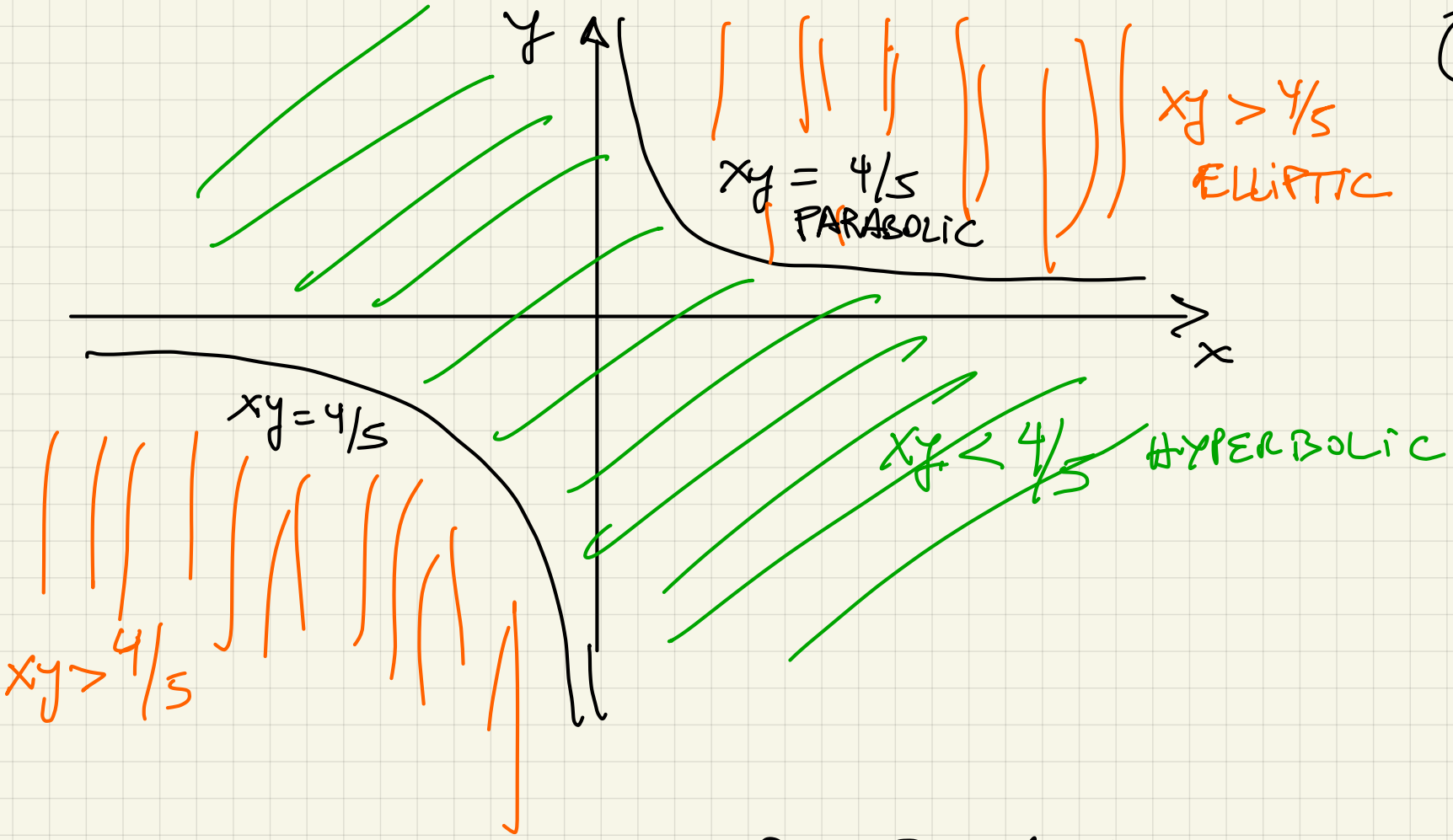
ELLIPTIC.

$$\Delta = 10xy - 8 < 0 \iff 5xy - 4 < 0 \iff \boxed{xy < 4/5}$$

HYPERBOLIC

$$\Delta = 10xy - 8 = 0 \iff 5xy - 4 = 0 \iff \boxed{xy = 4/5}$$

$\longrightarrow$ PARABOLIC



Where this classification comes from? So let's consider the case $\Delta = a_{22}a_{11} - a_{12}^2 > 0$ (<u>Elliptic Case</u>).

Goal is to make a change of variables so that we can view the equation as a Laplace's equation starting

from (†) in pg. 3 of L7b. $(x,y) \longrightarrow (x', y')$

Let
$$\left.\begin{aligned} x &= \alpha x' + \beta y' \\ y &= \gamma x' + \delta y' \end{aligned}\right\} \rightarrow \left\{\begin{aligned} \frac{\partial u}{\partial x'} &= \frac{\partial u}{\partial x}\cdot\alpha + \frac{\partial u}{\partial y}\cdot\gamma \\ \frac{\partial u}{\partial y'} &= \frac{\partial u}{\partial x}\beta + \frac{\partial u}{\partial y}\cdot\delta \end{aligned}\right.$$

by chain rule

So we have
$$\left\{\begin{aligned} u_{x'} &= \alpha u_x + \gamma u_y \\ u_{y'} &= \beta u_x + \delta u_y \end{aligned}\right.$$

We would like now that

$$L = \underbrace{\left(\partial_x + \frac{a_{12}}{a_{11}}\partial_y\right)^2}_{} + \overbrace{\left(\frac{a_{22}}{a_{11}} - \frac{a_{12}^2}{a_{11}^2}\right)}^{>0}\partial_{yy} =$$

$$L = \underbrace{\left(\partial_x + \frac{a_{12}}{a_{11}}\partial_y\right)^2}_{(\partial_{x'})^2} + \underbrace{\left[\left(\frac{a_{22}}{a_{11}} - \frac{a_{12}^2}{a_{11}^2}\right)^{1/2}\partial_y\right]^2}_{(\partial_{y'})^2}.$$ Thus

WANT IS TO CHOOSE $\alpha, \gamma, \beta, \delta$ so that

$$\begin{cases} u_{x'} = \alpha u_x + \gamma u_y = u_x + \dfrac{a_{12}}{a_{11}} u_y \\ u_{y'} = \beta u_x + \delta u_y = \left(\dfrac{a_{22}}{a_{11}} - \dfrac{a_{12}^2}{a_{11}^2}\right)^2 \cdot u_y \end{cases}$$

This means that $\alpha = 1$ $\gamma = \dfrac{a_{12}}{a_{11}}$ $\beta = 0$; $\delta = \left(\dfrac{a_{22}}{a_{11}} - \dfrac{a_{12}^2}{a_{11}^2}\right)^{1/2}$

Hence all in all we get that our equation is

$$0 = Lu = (\partial_{x'}^2 + \partial_{y'}^2) u.$$ which is a Laplace eq. in the new variables.

Similarly if $\Delta < 0$ then we would get.

$$0 = Lu = (\partial_{x'}^2 - \partial_{y'}^2) u$$ which is a wave equation in the new variables

If $\Delta = 0$ we'd get $0 = Lu = \partial_{x'}^2 u$ + lot $\to u_t$ <u>Parabolic</u>

<u>In higher dimensions $d \geq 2$</u> then

$$Lu = \underbrace{\sum_{i,j} a_{ij} u_{x_i x_j}}_{\text{all the second order terms.}} + \underbrace{\sum_i a_i u_{x_i} + a_0 u}_{\text{l.o.t}} = 0$$

Suppose $A = (a_{ij})_{\substack{1 \leq i \leq d \\ 1 \leq j \leq d}}$ is a <u>symmetric matrix</u> and <u>real valued</u>.

Fact from <u>linear algebra</u> A can be diagonalized; in fact one can find an orthogonal matrix B such that

$$BA\,{}^tB = \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_d \end{pmatrix}$$

$\lambda_1, \lambda_2, \ldots \lambda_d$ are called the EIGENVALUES OF A.

${}^tB$ = transpose of B $\quad B = (b_{ij})$ $\quad {}^tB = (b_{ji})$

Now the classification is as follows:

(1) L is <u>ELLIPTIC</u> iff A or $(-A)$ is positive definite; that is all eigenvalues of A or of $(-A)$ are <u>positive</u>. (>0).

(2) L is <u>Hyperbolic</u> iff A or $(-A)$ has <u>ALL BUT</u> <u>ONE</u> positive eigenvalues.

NEGATIVE.

(3) L is <u>PARABOLIC</u> iff A has one zero eigenvalue and the rest have all the same sign.

When $d=2$ which is what we did before this translates to:

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$\boxed{\Delta = a_{22}a_{11} - a_{12}^2}$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \longleftrightarrow \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$\Delta > 0$ (ELLIPTIC) $\Longleftrightarrow$ λ_1, λ_2 have the same sign

$\Delta < 0$ (Hyperbolic) $\Longleftrightarrow$ λ_1, λ_2 have different signs (~~non zero~~).

$\Delta = 0$ (PARABOLIC) $\Longleftrightarrow$ $\lambda_1 = 0$, $\lambda_2 \neq 0$.

The (1+1) Linear Wave EQUATION (hyperbolic).

$x \in \mathbb{R}$, $t \in \mathbb{R}$; it will be enough to consider $t > 0$ because we'll later that the wave equation is time reversible.

$u(x,t) \rightleftarrows u(x,-t)$ (forward or backwards in time)

$$\dagger \qquad u_{tt} - c^2 u_{xx} = 0 \qquad c = \text{wave speed constant}$$

$$(\partial_{tt} - c^2 \partial_{xx}) u = 0 \qquad L = \partial_{tt} - c^2 \partial_{xx} \Longrightarrow$$

$$L = (\partial_t + c\partial_x)(\partial_t - c\partial_x) =: \Box$$

So the wave equation (+) can be rewritten as $\Box u = 0$

and as

$$(\partial_t + c\partial_x)\underbrace{(\partial_t - c\partial_x)u}_{=:\, v(x,t)} = 0 \qquad u(x,t)$$

① $\underline{(\partial_t + c\partial_x)}v = 0$ → homog. transport.

first. →

② and then solving u from v. $\boxed{(\partial_t - c\partial_x)u = v}$

inhomog. transport

To solve ① we know that

① $\boxed{v(x,t) = g(x-ct)}$ where g is any differentiable

function of one variable $g: \mathbb{R} \to \mathbb{R}$, since the characteristic lines of $(\partial_t + c\partial_x)v = 0$ are $x - ct = \text{const}$.

Now we need to solve the inhomogeneous transport equation

(2) $$(\partial_t - c\partial_x)u = v \quad \text{with } v \text{ as above}$$

$$\Longleftrightarrow \quad \boxed{u_t - c u_x = g(x-ct)}$$

Need to solve this to find u.

How do we do it? We've seen this in L6-L7a)!

(continue next time).